

BRAC University (Department of Computer Science and Engineering)
CSE 260 (Digital Logic Design)
Summer 2025
Set A

Examination: Quiz 3

Duration: 25 Minutes

Full Marks: 15

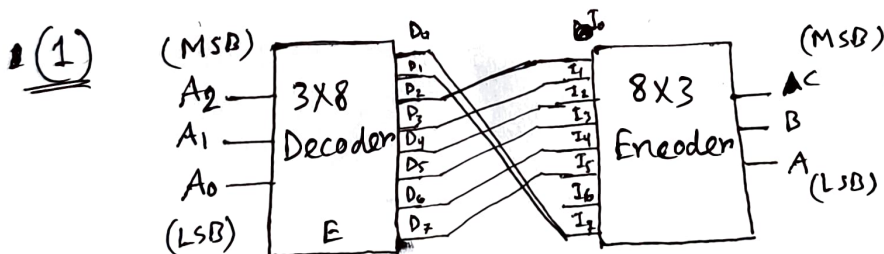
Answer the following questions. You **MUST** show your workings/calculations where applicable.

Name:

ID:

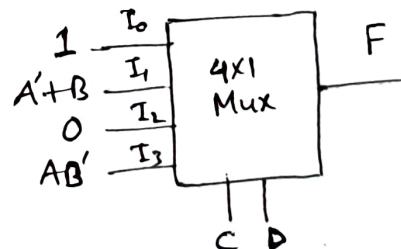
Section:

1. Consider A is a 3-bit number. Design a circuit that performs A - 2 using encoder and decoder. In case the answer is less than 0, the output should be 7. [5]
2. Implement the following function using a 4:1 Mux. [5]
 $F(A, B, C, D) = \sum(0, 1, 4, 5, 8, 11, 12, 13)$
3. Consider A is a 5-bit number. Design A-3 (A minus 3) using a 5-bit parallel adder. Use external gates if required. [5]



(2)

	I_0	I_1	I_2	I_3
$A'B'$	0	1	2	3
$A'B$	4	5	6	7
AB'	8	9	10	11
AB	12	13	14	15



$$\begin{aligned}
 I_0 &= 1 \\
 I_1 &= A'B' + A'B + AB \\
 &= A'(B' + B) + AB \\
 &= A' + AB \\
 &= (A' + A)(A' + B) = A' + B \\
 I_2 &= 0 \\
 I_3 &= AB'
 \end{aligned}$$

(3)

